Expno:12 End - End Communication at Transport Layer.

7.10.26

Aim:

To implement End - End communication at Transport Layer.

a. Implement echo client server using TCP/UDP sockets

TCP :-

```
import socket
import threading
# SERVER
def start_server():
    server = socket.socket(socket.AF_INET
        socket.SOCK_STREAM)
server.bind('127.0.0.1',5000))
server.listen(1)
print("[SERVER] Listening on 127.0.0.1:5
conn, addr = server.accept()
print(f'[SERVER] Connected by [addr]")
while True:
    data = conn.recv(1024).decode()
if not data or data.lower() == 'quit':
    print(f'[SERVER] Receives : {data}")
conn.close()
server.close()
```

CLIENT

```
def start_client():
    client = socket.socket(socket.AF_INET,
        socket.SOCK_STREAM)
    client.connect(('127.0.0.1', 50001))
    print("[CLIENT]")
    while True:
        msg = input("Enter Message")
        client.send(msg.encode())
        if msg.lower() == 'quit':
            break
    print(f"[SERVER RESPONSE]{response}")
    client.close()
```

MAIN

```
if __name__ == "__main__":
    server_thread = threading.Thread
        (target = start_server,
    daemon=True)
    server_thread.start()
    start_client()
```

OUTPUT:-

[SERVER] Listening on 127.0.0.1:5000
[CLIENT] Connected to server.
[SERVER] Connected by ('127.0.0.1', 53458) Enter message:
hello
[SERVER] Received: hello
[SERVER] Listening on 127.0.0.1 5000
[CLIENT] Connected to server.
[SERVER] Connected by ('127.0.0.1', 60328) Enter message:
Hi
[SERVER] Received: Hi
[SERVER RESPONSE] Server Received: Hi
Enter Message: quit
Connection closed

UDP:

```
import socket
import threading
def udp_server():
    s = socket.socket(socket.AF_INET, socket)
    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
```

```
while True:
    data, addr = s.recvfrom(1024)
    msg = data.decode()
    if msg.lower() == 'quit':
        print(f"SERVER] Received
from {addr}: {msg}")
    s.close()
    print("[SERVER] stopped.")

#CLIENT
def udp_client():
    c=socket.socket(socket.AF_INET,
socket.SOCK_DGRAM)
    server=('127.0.0.1', 12000)
while TRUE:
    msg = input("Enter message")
    c.sendto(msg.encode(), server)
    if msg.lower() == 'quit':
        print("[CLIENT] closing
connection")
        break
    c.close()
    print("[CLIENT] stopped").
```

```
# - MAIN -
if __name__ == '__main__':
    threading.Thread(target=udp_server,
    daemon=True).start()udp_client()
```

OUTPUT:

```
Enter message :
[SERVER] Received from ('127.0.0.1'):
    hi
Enter message : hello
[SERVER] Received from ('127.0.0.1',
49670): how are you
Echo from server : how are you
Enter message : quit
[CLIENT] closing connection
[SERVER] client ('127.0.0.1', 49670)
ended connection
[CLIENT] Stopped [SERVER] Stopped
```

Result:

Thus, the implementation of end-end communication is done successfully.